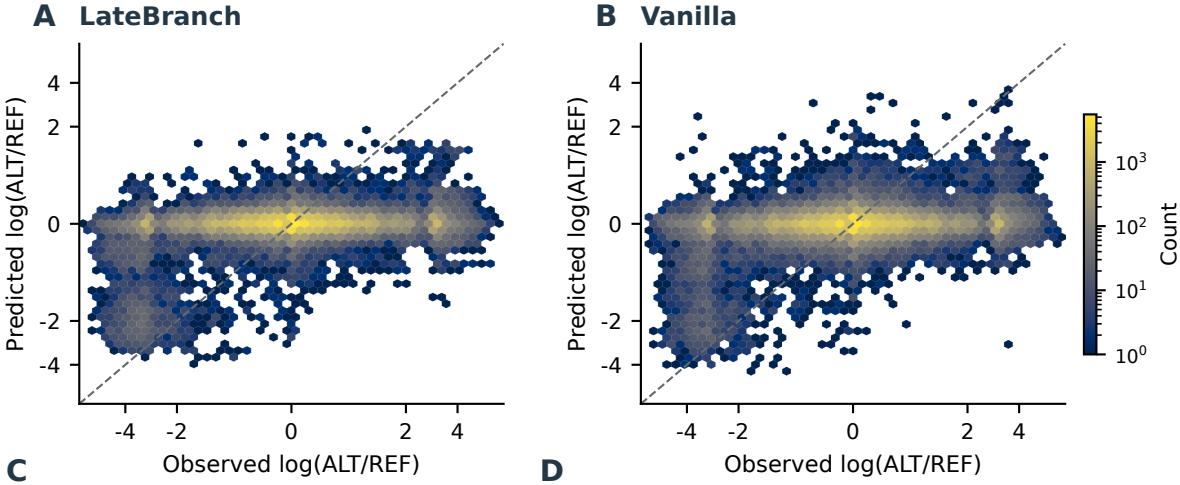


C

WINDOW HOLDOUT

Seen chromosomes · new loci

Train	chr1–22	680,404
Val	chr1–22	98,019
Cal	chr1–22	95,734
Test	chr1–22	97,662

257-bp windows · zero cross-split overlap

D

CHROMOSOME HOLDOUT

Unseen chr19–22 at test time



Train 621,836 · Val 68,854 · Cal 31,145 · Test 249,984

Test chromosomes excluded from all earlier stages

E

INDEPENDENT CALIBRATION

Fixed raw log-ratio τ → Fit s_a on cal counts → **Test** $p_{ALT} = \sigma(s_a \tau)$

LB scales: window 0.57–0.81 · chr 0.08–0.29 · training ρ_a fixed

F

LATEBRANCH: LOWER TEST NLL AFTER CALIBRATION

	RAW		CALIBRATED	NLL LOWER BY
Window	2.76675	→	2.76561	**1.14×10^{-3}**
chr19–22	2.73013	→	2.71755	**12.57×10^{-3}**